Modern Algebra 1

Grant Talbert
09/26/24
Dr. Duque-Rosero
MA 541 Section A1

Boston University

Problem 1

Solution. For convenience, let I always denote the 2×2 identity matrix within this problem, and let $H \coloneqq \left\{\alpha I \mid \alpha \in \mathbb{Z}_n \wedge \gcd(n, a^2) = 1\right\}$.

First, we show that $H \subseteq Z(\mathrm{GL}_2(\mathbb{Z}_n))$. Let $A \in \mathrm{GL}_2(\mathbb{Z}_n)$, and let $\alpha \in \mathbb{Z}_n$. It follows that

$$(\alpha I)A = \alpha(IA) = \alpha A = A\alpha = (AI)\alpha = A(I\alpha) = A(\alpha I).$$

Thus, for all $A \in \mathrm{GL}_2(\mathbb{Z}_n)$, $\alpha IA = A\alpha I$. Therefore, $H \subseteq Z(\mathrm{GL}_2(\mathbb{Z}_n))$.

Now, we show the converse. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad B = \begin{bmatrix} e & f \\ g & h \end{bmatrix}.$$

It follows that

$$AB = \begin{bmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{bmatrix}, \qquad BA = \begin{bmatrix} ae + cf & be + df \\ ag + ch & bg + dh \end{bmatrix}.$$

We want $AB = BA$. This is true if and only if

$$\begin{aligned} ae + bg &= ae + cf \\ af + bh &= be + df \\ ce + dg &= ag + ch \\ cf + dh &= bg + dh. \end{aligned}$$

Since the only terms present on both sides of an equation are ae and dh, this will **not** be true in general unless $b, c = 0$. We then have

$$\begin{bmatrix} ae & af \\ dg & dh \end{bmatrix} = \begin{bmatrix} ae & df \\ ag & dh \end{bmatrix}.$$

From this, it follows that $ag = dg$, and thus $a = d$. Therefore, $A = \alpha I$, and $Z(\mathrm{GL}_2(\mathbb{Z}_n) \subseteq H$.

Therefore, $Z(\mathrm{GL}_2(\mathbb{Z}_n) = H$.

□

Problem 2

Solution. This was easy, in fact we did an example of this last homework. Let $G = \{e, a, b, c\}$ be a group with identity e. Let $a^2 = b^2 = c^2 = e$, and let $ab = ba = c$. It follows that $bc = cb = a$ and $ac = ca = b$. We give the following Cayley table to help with this visualization.

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

An equivalent group up to isomorphism is the subgroup $\{R_0, R_{180}, S, R_{180}S\}$ of the dihedral group D_n for an even integer n.

The group is clearly not cyclic. It has 4 elements, none of which generate the group. Since $a^2 = b^2 = c^2 = e$, we have

$$\begin{aligned} \langle a \rangle &= \{e, a\}, \\ \langle b \rangle &= \{e, b\}, \\ \langle c \rangle &= \{e, c\}. \end{aligned}$$

Therefore, the set is not cyclic. Additionally, these sets and the set $\{e\}$ are the only possible cyclic subgroups of G, which is obvious since they are the sets generated by each element of G. It remains to be shown no other subgroups of G exist.

Let $H \subsetneq G$ have more than 2 elements. Since any set must have the identity element in it, we have already seen all the subgroups of G with 2 or less elements. They are exactly the cyclic subgroups of G. We must only consider sets with more than 2 but less than 4 elements. In other words, we must consider sets with only 3 elements. All of these sets, in order to be groups, must have the identity element, so they must have exactly 2 non-identity elements a, b, or c in them. Since $ab = ba = c$, $bc = cb = a$, and $ca = ac = b$, none of these sets would be closed under group multiplication. Thus, there exist no proper subgroups of G other than the cyclic subgroups. $\square$

Problem 3

Let $\langle a \rangle$ a group with order n, and let $d|n$. Show there are $\phi(d)$ elements of a with order d.

Solution. The fundamental theorem of finite cyclic groups guarentees for every divisor d of n, there exists a unique group $\left\langle a^{n/d} \right\rangle$ of order d. We must show that this subgroup has $\phi(d)$ generators. Let k be relatively prime to d, and notice that $a^{(n/d)k} \in \left\langle a^{n/d} \right\rangle$. Since the order of this group is d, we have the order of $a^{(n/d)k}$ in $\left\langle a^{n/d} \right\rangle$ is

$$\left| a^{(n/d)k} \right| = \frac{d}{\gcd(d, k)}.$$

Since $\gcd(d, k) = 1$, we have $\left| a^{(n/d)k} \right| = d$. Therefore, for any k relatively prime to d, there exists an element $a^{(n/d)k}$ with order d. Therefore, the number of elements with order d are simply the number of numbers relatively prime to d; that is, $\phi(d)$. $\square$

Problem 4

Solution. Let G be a group with identity e. Let $a, b \in G$ such that $|a| = n$, $|b| = m$, and $\gcd(n, m) = 1$.

Let there exist some $g \in \langle a \rangle$ such that $g \in \langle b \rangle$. Since $g \in \langle a \rangle$, there must exist some $k \in \mathbb{Z}$ such that $g = a^k$. Since $\langle b \rangle$ is a group and thus closed under multiplication, any power of a^k must be an element of $\langle b \rangle$. Thus, we have

$$\left\langle a^k \right\rangle \subseteq \langle b \rangle.$$

By the fundamental theorem of cyclic subgroups, we know that $\left|\left\langle a^k \right\rangle\right|$ will divide m. Additionally, by the same logic $\left\langle a^k \right\rangle \subseteq \langle a \rangle$, so $\left|\left\langle a^k \right\rangle\right|$ must also divide n. Since m and n are relatively prime, the only number that divides both m and n is 1, and thus $\left|\left\langle a^k \right\rangle\right| = 1$. It follows from this that $a^k = e$. Therefore, $g \in \langle a \rangle$ and $g \in \langle b \rangle$ implies $g = e$. This should suffice to show $\langle a \rangle \cap \langle b \rangle = \{e\}$. $\square$

Problem 5

Solution. Let $a \in G$. The set $C(a)$ is the set of all $g \in G$ such that $ga = ag$. We need only show $\langle a \rangle \subseteq C(a)$. Since $a^k \in \langle a \rangle$ for any $k \in \mathbb{Z}$, we have

$$a\left(a^k\right) = a\left(a^{k-1}a\right) = \left(aa^{k-1}\right)a = a^k a.$$

Therefore, for any $a^k \in \langle a \rangle$, $a^k a = aa^k$. Thus, $\langle a \rangle \subseteq C(a)$. $\square$